CHAPTER 11

INTERNATIONAL TRADE

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1 INTRODUCTION

A country does not produce all the goods and services that it consumes. Instead, it produces the goods and services in which it is good at producing and imports the goods and services in which it is not good at producing. International trade is the exchange of goods and services across international borders. This chapter gives an exposition of international trade.

2 BASIS OF SPECIALISATION AND INTERNATIONAL TRADE

2.1 Law of absolute advantage

The father of modern economics, Adam Smith, was the first economist who recognized and advocated the importance and the gains from specialisation and international trade. He put forward the law of absolute advantage in his famous book, "The Wealth of Nations", which was published in 1776.

According to the law of absolute advantage, countries can gain from specialisation and international trade if each specialises in producing the goods in which it has an absolute advantage. A country has an absolute advantage over other countries in producing a good when it can produce the same amount of the good with a smaller amount of resources.

Suppose that there are two countries, country X and country Y, producing two goods, good A and good B. Further suppose that there are perfect mobility of resources within each country, constant opportunity costs of production, no economies of scale, no transport costs, no trade barriers and no product differentiation.

	Good A	Good B
Country X	2	4
Country Y	1	9

The above table shows the amount of each good that can be produced in each country with one unit of resources. In country X, one unit of resources can be used to produce either 2 units of good A or 4 units of good B. In country Y, one unit of resources can be used to produce either 1 unit of good A or 9 units of good B.

In country X, 1/2 unit of resources is required to produce one unit of good A and 1/4 unit of resources is required to produce one unit of good B. In country Y, 1 unit of resources is required to produce one unit of good A and 1/9 unit of resources is required to produce one unit of good B. Since the amount of resources required to produce one unit of good A in country X is lower than that in country Y and the amount of resources required to produce one unit of good B in country Y is lower than that in country X, country X has an absolute advantage in producing good A and country Y has an absolute advantage in producing good B.

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Suppose that country X has 400 units of resources and country Y has 200 units of resources. Further suppose that each country allocates its resources equally between the two goods.

	Good A	Good B
Country X	400	800
Country Y	100	900
World	500	1700

The above table shows the amount of each good produced in each country when the resources are allocated equally between the two goods. In country X, 400 units of good A and 800 units of good B are produced. In country Y, 100 units of good A and 900 units of good B are produced.

Suppose that each country completely specialises in producing the good in which it has an absolute advantage.

	Good A	Good B
Country X	800	0
Country Y	0	1800
World	800	1800

The above table shows the amount of each good produced in each country when each country completely specialises in producing the good in which it has an absolute advantage. In country X, 800 units of good A are produced. In country Y, 1800 units of good B are produced. The world output of good A has increased by 300 units and the world output of good B has increased by 100 units.

Consider what will happen if the two countries trade. Suppose that that country X trades 250A for 850B.

	Good A	Good B
Country X	550	850
Country Y	250	950
World	800	1800

The above table shows the amount of each good available for consumption in each country after specialisation and international trade. In country X, 550 units of good A and 850 units of good B are available for consumption. In country Y, 250 units of good A and 950 units of good B are available for consumption.

Since the amount of each good in each country available for consumption has increased, we can conclude that countries can gain from specialisation and international trade on the basis of the law of absolute advantage.

Note: Although the law of absolute advantage is no longer in the syllabus, it is useful for understanding the law of comparative advantage.

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2.2 Law of comparative advantage

In the previous section, country X has an absolute advantage in producing good A and country Y has an absolute advantage in producing good B. The next natural question is, "If one of the two countries has an absolute advantage in producing both goods, will the two countries still gain from specialisation and international trade? This is the question that David Ricardo asked, after reading "The Wealth of Nations", which was published in 1776 by Adam Smith. He answered the question in the affirmative and put forward the law of comparative advantage in his famous book, "On the Principles of Political Economy and taxation", which was published in 1817.

According to the law of comparative advantage, countries can gain from international trade if each specialises in producing the goods in which it has a comparative advantage. A country has a comparative advantage over other countries in producing a good when it can produce the same amount of the good at a lower opportunity cost. In other words, it can produce the same amount of the good by forgoing a smaller amount of other goods.

Suppose that there are two countries, country X and country Y, producing two goods, good A and good B. Further suppose that there are perfect mobility of resources within each country, constant opportunity costs of production, no economies of scale, no transport costs, no trade barriers and no product differentiation.

	Good A	Good B
Country X	2	4
Country Y	3	9

The above table shows the amount of each good that can be produced in each country with one unit of resources. In country X, one unit of resources can be used to produce either 2 units of good A or 4 units of good B. In country Y, one unit of resources can be used to produce either 3 units of good A or 9 units of good B. Although country Y has an absolute advantage in producing both goods, each country has a comparative advantage in producing only one good.

In country X, if one unit of resources is used to produce 2 units of good A, the same unit of resources cannot be used to produce 4 units of good B. Therefore, the opportunity cost of producing 1 unit of good A in country X is 2 units of good B. By the same token, the opportunity cost of producing 1 unit of good B in country X is 1/2 unit of good A. In country Y, if one unit of resources is used to produce 3 units of good A, the same unit of resources cannot be used to produce 9 units of good B. Therefore, the opportunity cost of producing 1 unit of good A in country Y is 3 units of good B. By the same token, the opportunity cost of producing 1 unit of good B in country Y is 1/3 unit of good A. Since the opportunity cost of producing good A in country X is lower than that in country Y and the opportunity cost of producing good B in country Y is lower than that in country X, country X has a comparative advantage in producing good A and country Y has a comparative advantage in producing good B.

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Suppose that country X has 400 units of resources and country Y has 200 units of resources. Further suppose that each country allocates its resources equally between the two goods.

	Good A	Good B
Country X	400	800
Country Y	300	900
World	700	1700

The above table shows the amount of each good produced in each country when the resources are allocated equally between the two goods. In country X, 400 units of good A and 800 units of good B are produced. In country Y, 300 units of good A and 900 units of good B are produced.

Suppose that each country completely specialises in producing the good in which it has a comparative advantage.

	Good A	Good B
Country X	800	0
Country Y	0	1800
World	800	1800

The above table shows the amount of each good produced in each country when each country completely specialises in producing the good in which it has a comparative advantage. In country X, 800 units of good A are produced. In country Y, 1800 units of good B are produced. The world output of good A and the world output of good B have each increased by 100 units.

Consider what will happen if the two countries trade.

In country X, the opportunity cost of producing 1 unit of good B is 1/2 unit of good A. Therefore, country X will only be willing to specialise completely in producing good A and trade if 1/2 unit of good A can be exchanged for more than 1 unit of good B (1B < 1/2A) because in this range of exchange ratios, the opportunity cost of buying 1 unit of good B from country Y is less than 1/2 unit of good A. In country Y, the opportunity cost of producing 1 unit of good A is 3 units of good B. Therefore, country Y will only be willing to specialise completely in producing good B and trade if 3 units of good B can be exchanged for more than 1 unit of good A (1A < 3B) because in this range of exchange ratios, the opportunity cost of buying 1 unit of good A from country X is less than 3 units of good B. Therefore, the range of mutually beneficial exchange ratios is 2B < 1A < 3B. The actual exchange ratio depends on the demand and the supply of the two goods and their elasticities of demand and supply in the two countries.

Suppose that given the demand and supply of the two goods and their elasticities of demand and supply in the two countries, the exchange ratio is 1A = 2.5B. Further suppose that country X trades 350A for 875B (350 x 2.5).

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	Good A	Good B
Country X	450	875
Country Y	350	925
World	800	1800

The above table shows the amount of each good available for consumption in each country after specialisation and international trade. In country X, 450 units of good A and 875 units of good B are available for consumption. In country Y, 350 units of good A and 925 units of good B are available for consumption.

Since the amount of each good in each country available for consumption has increased, we can conclude that countries can gain from specialisation and international trade on the basis of the law of comparative advantage.
The gains from specialisation and international trade can be illustrated with a diagram.

Country X Country Y

In the above diagrams, without specialisation and international trade, the Consumption Possibility Curve (CPC) is the same as the Production Possibility Curve (PPC) in each country. With specialisation and international trade, the CPC lies to the right of the PPC in each country. Therefore, specialisation and international trade is beneficial.

Note: Countries do not gain equally from specialisation and international trade, unless by chance. The closer the exchange ratio to the pre-trade exchange ratio in a country, the smaller the gains from specialisation and international trade to the country. The converse is also true.

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2.3 Limitations of the law of comparative advantage

Trade barriers

The law of comparative advantage assumes that there are no trade barriers. In reality, there are trade barriers. Some governments protect domestic industries by discouraging imports through the use of protectionist measures such as tariffs, import quotas and subsidies. As a result, some countries produce goods in which they do not have a comparative advantage. Further, some countries are unable to export the goods in which they have a comparative advantage as much as they would like due to trade barriers imposed by foreign governments. For instance, Singapore has a comparative disadvantage in producing drinkable water due to the small land area and hence limited reservoirs and water catchment areas. However, to reduce the country's dependence on imported water, the Singapore government subsidises the production of drinking water through seawater desalination and wastewater reclamation to increase self-sufficiency in its water supply.

Product differentiation

The law of comparative advantage assumes that there is no product differentiation. In reality, differentiated goods are produced. Further, due to factors such as intra-industry specialisation, a country does not produce all types of a good. For instance, Singapore's top two exports are electronic valves and refined petroleum products. However, they are also Singapore's top two imports. This is partly because Singapore imports and exports different types of electronic valves and refined petroleum products.

Transport costs

The law of comparative advantage assumes that there are no transport costs. In reality, there are transport costs which may outweigh any comparative advantage or disadvantage. For instance, Singapore has a comparative disadvantage in producing bricks due to the small amount of low-skilled labour. However, it produces bricks because their size and weight make them too expensive to import.

Constant opportunity costs of production

The law of comparative advantage assumes that there are constant opportunity costs of production. In reality, as a country increasingly specialises in producing a good, it will experience increasing opportunity cost of producing the good. This is because factor inputs are not equally suitable for producing different goods. As a country increasingly specialises in producing a good, it has to use resources that are less suitable for producing the good to actually produce the good. This means that increasingly more units of resources are needed to produce each additional unit of the good. Therefore, increasingly more units of other goods have to be given up to produce each additional unit of the good. This will eventually lead to the disappearance of the country's comparative advantage in producing the good which is a reason why countries do not engage in complete specialisation in reality.

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Perfect mobility of resources within each country
The law of comparative advantage assumes that there is perfect mobility of resources within each country. In reality, due to factors such as differences in skill requirements, resources are not perfectly mobile in a country.

Supply-side reason
The law of comparative advantage does not take into consideration the demand-side reason for international trade. In reality, countries also trade due to differences in demand conditions. Refer to section 2.5.

Sources of comparative advantage
The law of comparative advantage does not provide an explanation for the sources of comparative advantage.

2.4 Sources of comparative advantage

Differences in factor endowments
Factor endowments vary among countries. For example, Argentina has much fertile land, Saudi Arabia has large crude oil reserves and China has a large pool of unskilled labour. Since goods differ according to the resources that are used to produce them, a country has a comparative advantage in producing goods that intensively use resources it has in abundance. For example, Argentina has a comparative advantage in growing wheat because of its abundance of fertile land, Saudi Arabia has a comparative advantage in producing oil because of its abundance of crude oil reserves and China has a comparative advantage in producing textile because of its abundance of unskilled labour.

Economies of scale
A country may acquire a comparative advantage in producing a good through large-scale production. This occurs in industries where production is subject to economies of scale.

2.5 Demand-side reason for international trade

The law of comparative advantage provides the supply-side reason for international trade. However, in addition to differences in supply conditions, international trade also takes place due to differences in demand conditions.

Suppose that Singapore and Japan have identical concave PPC and hence identical supply curve for fish which means that neither country has a comparative advantage over the other in producing fish. If the demand for fish is higher in Singapore than in Japan, the price of fish in Singapore will be higher than that in Japan. Therefore, Singapore will import fish from Japan. If the demand for fish is higher in Japan than in Singapore, the price of fish in Japan will be higher than that in Singapore. Therefore, Japan will import fish from Singapore. In these cases, international trade will take place due to difference in demand conditions.

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When this happens, the world price of fish, which will be determined by the world demand and the world supply of fish, will be below the equilibrium price in Singapore and above the equilibrium price in Japan. Therefore, Singapore will benefit from importing fish from Japan at a lower price.

In the above diagram, the world price of fish (P_W) is below the equilibrium price in Singapore (P_S) and higher than the equilibrium price in Japan (P_J). At P_W, the shortage of fish in Singapore equals the surplus of fish in Japan, which means that the world demand for fish equals the world supply of fish. Similarly, Japan will benefit from importing goods with a higher demand in the country from Singapore.

2.6 Advantages of international trade

Higher consumption (which has been explained with the law of comparative advantage)
International trade increases the amount of goods available to the people in a country for consumption. In other words, international trade shifts the Consumption Possibility Curve (CPC) of a country outwards.

Greater variety of goods
Certain goods cannot be produced in a country due to lack of certain resources. Therefore, international trade increases the variety of goods available to the people in a country for consumption.

Lower prices
International trade allows consumers in a country to buy goods at lower prices from foreign firms that are more efficient than domestic firms.

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Increased efficiency
International trade increases competition and hence drives domestic firms to increase efficiency. Faced with competition from imports, domestic firms have to be more efficient to survive.

Specialisation
Without international trade, a country would not specialise in producing the goods in which it had a comparative advantage due to the small sizes of the domestic markets for the goods. Further, the absence of international trade would compel a country to produce goods in which it had a comparative disadvantage.

Technological transfer
International trade allows a country to gain access to modern capital equipment such as computers which will lead to higher productivity in the country. In other words, international trade shifts the Production Possibility Curve (PPC) of a country outwards.

2.7 Pattern of trade between Singapore and the rest of the world

The pattern of trade between Singapore and the rest of the world refers to the types of goods that Singapore imports and exports. It is determined by three factors: supply factors, demand factors and government policies.

Singapore's exports and imports are affected by supply factors
Singapore used to have a comparative advantage in producing low value-added goods due to the large amount of low-skilled labour. Therefore, Singapore' main exports were low-value-added goods such as semiconductors and disk drives, and to a lesser extent, television sets, radios, clothing and plastics. However, over the last few decades, the skill level of the labour force in Singapore has been rising due to the great emphasis on education and training, the immigration policy and the foreign worker policy. Singapore now has a comparative advantage in producing high value-added goods due to the large amount of high-skilled labour. Therefore, Singapore's main exports are high value-added goods which include high-end electronics such as electronic valves, and high-end chemicals such as refined petroleum products, and to a lesser extent, engineering products such as civil engineering equipment parts and pharmaceuticals. Singapore also exports high value-added services such as financial services, education and healthcare. Although Singapore continues to export low value-added goods such as semiconductors and disk drives, the share of these goods in Singapore's exports has decreased. Singapore virtually does not have factor endowments and hence has a comparative disadvantage in extracting natural resources such as crude petroleum. Further, Singapore has limited fertile land and hence has a comparative disadvantage in agriculture. Therefore, Singapore imports crude petroleum and agricultural products. Due to the small amount of low-skilled labour, Singapore also has a comparative disadvantage in producing low value-added goods such as textile and furniture which results in the import of these goods.

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Singapore's exports and imports are affected by demand factors
Suppose that Singapore and Japan have identical concave PPC and hence identical supply curve for fish which means that neither country has a comparative advantage over the other in producing fish. If the demand for fish is higher in Singapore than in Japan, the price of fish in Singapore will be higher than that in Japan. Therefore, Singapore will import fish from Japan. If the demand for fish is higher in Japan than in Singapore, the price of fish in Japan will be higher than that in Singapore. Therefore, Japan will import fish from Singapore. In these cases, international trade will take place due to difference in demand conditions.

Singapore's exports and imports are affected by government policies
The Singapore government has provided infrastructures such as Jurong Island for chemical manufacturing and Biopolis for pharmaceutical manufacturing and given incentives such as grants and tax concessions to induce foreign chemical firms and pharmaceutical firms to invest in Singapore. As a result, Singapore has developed a comparative advantage in producing chemicals and pharmaceuticals. Singapore has signed close to 20 free trade agreements and this has also led to an increase in the foreign direct investments enabling it to develop a comparative advantage in some high value-added industries. To reduce the Singapore's dependence on imported water, the government subsidises the production of drinking water through seawater desalination and wastewater reclamation to increase self-sufficiency in its water supply.

3 PROTECTIONISM

Although international trade is beneficial, free trade may be undesirable which gives rise to the arguments for protectionism. Protectionism is the use of protectionist measures and hence is a departure from free trade with the purpose of protecting domestic industries from foreign competition.

3.1 Protectionist measures

Tariffs
Tariffs are taxes imposed on imports and they can be used to increase the price of imports to decrease the quantity.

Import quotas
Import quotas are limits imposed on the quantities of certain imports and they can be used to decrease the quantity of imports.

Subsidies
Subsidies can be given to domestic firms to reduce their cost and hence price. They can be given to domestic firms which produce goods that compete with imports to decrease the quantity of imports. They can also be given to domestic firms which produce goods for export to increase the quantity of exports.

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Procurement policies
The government can adopt a policy of buying goods and services from domestic firms to decrease the quantity of imports even if they are more expensive or of lower quality than those produced by foreign firms.

Voluntary export restraints
Voluntary export restraints are agreements between two economies where the government of the exporting economy agrees to limit the quantities of certain goods exported to the importing economy. They are usually signed in the face of threatened actions by the government of the importing economy. For instance, Japan has entered into a number of voluntary export restraints with EU members and with the USA in the export of its cars.

Exchange controls
Exchange controls are limits on foreign currency made available to domestic residents and they can be used to decrease the quantity of imports.

Health and safety regulations
Health and safety standards of imports can be used to decrease the quantity of imports.

Embargoes
Embargoes are bans on certain imports and they can be used to decrease the quantity of imports.

3.2 Arguments for and against protectionism

Infant industry argument (Sunrise industry argument)
Protectionism allows some new and small domestic industries to grow and hence develop a comparative advantage which will help them compete with mature and big foreign industries. When an industry expands, it will reap more economies of scale which will lead to a fall in the average cost of production and this allows it to compete with fully-fledged foreign rivals. Until then, the industry may need to be protected to survive its infancy. However, this argument is often criticised on three grounds. First, the government may not be able to identify the right industries that merit protection. Second, the very existence of protection may foster inefficiency which may lead to the outcome that the protected industries never realise the expected economies of scale and thus never become competitive. Third, once protection is given, it is often hard to remove.

Declining industry argument (Sunset industry argument)
Protectionism allows declining industries to decline less rapidly resulting in lower structural unemployment. The structure of the economy changes when some industries expand and some industries contract and this could be due to technological advancements, changes in comparative advantage or changes in the pattern of demand. When this happens, the expanding industries will create jobs and the contracting industries will lose jobs. However, as many of the workers who will lose their jobs in the contracting industries do not have the relevant skills and knowledge to find jobs in the expanding industries,

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structural unemployment will occur. To reduce structural unemployment, the government can give protection to the declining industries to allow them to decline less rapidly so that the workers in the industries have sufficient time to undergo education and training to acquire the relevant skills and knowledge required to find jobs in the expanding industries.

Job protection argument
Protectionism may protect jobs in a country in a recession. However, this argument is often criticized on two grounds. First, if the economy's trading partners retaliate, the rise in job opportunities in the protected industries may be offset by a fall in job opportunities in other industries. Second, even in the absence of retaliation, job opportunities in other industries may also fall if the trade barriers lead to a fall in the national income and hence the imports of the economy's trading partners.

Balance of payments argument
Protectionism may correct a persistent balance of payments deficit. However, this argument is often criticized on three grounds. First, if the cause of the persistent balance of payments deficit is high cost of production or low product quality, protectionism will not solve the underlying problem. Second, if the economy's trading partners retaliate, the fall in import expenditure may be offset by a fall in export revenue, leaving the current account balance unchanged. Third, even in the absence of retaliation, export revenue may also fall if protectionism leads to a fall in the national income and hence the imports of the economy's trading partners.

Anti-dumping argument
Protectionism is a countervailing measure against dumping. Dumping occurs when imports are sold in the domestic market at prices below their marginal costs, often as a result of foreign government subsidies. Although consumers will benefit from the lower prices, domestic firms may be driven out of the market. If this happens, once foreign firms monopolize the domestic market, consumers may suffer from high prices.

Diversification argument
A highly specialised country, such as Zambia with copper and Cuba with sugar, is rather susceptible to world market fluctuations, and protectionism provides greater diversity which reduces these risks.

Key industry argument (Strategic industry argument)
Some industries produce vital goods such as food, water and armaments, and protectionism helps a country maintain a certain degree of self-sufficiency in these areas.

In addition to the criticisms of the arguments discussed above, protectionism is also often criticized on other grounds. First, the immediate cost of increasing tariffs and imposing import quotas is higher prices for consumers and subsidies may become a strain on the budget of the government. Second, the gains from complete specialisation cannot be realised. Last but not least, protecting one stage of production often requires protecting downstream stages of production. For example, protecting the US textile industry from foreign competition may raise the cost of cloth to US garment manufacturers. Thus, if the

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government protects the domestic textile industry, it may also need to protect the domestic garment industry.

4 TERMS OF TRADE

The terms of trade are the number of units of imports that can be obtained with one unit of exports. It is expressed as the ratio of the price of exports to the price of imports.

$$\text{Terms of trade} = \frac{\text{Price of exports}}{\text{Price of imports}}$$

The terms of trade are often expressed as an index.

$$\text{Terms-of-trade index} = \frac{\text{Price index of exports}}{\text{Price index of imports}} \times 100$$

If the terms of trade rise (the price of exports rises relative to the price of imports), they are said to have improved or moved in a favourable direction. If the terms of trade fall (the price of exports falls relative to the price of imports), they are said to have deteriorated or moved in an unfavourable direction.

4.1 Factors affecting the terms of trade

Changes in the terms of trade may be due to three factors: demand factors, supply factors and the exchange rate.

Demand factors
An increase in the demand for exports will lead to a rise in the price and hence an improvement in the terms of trade. A decrease in the demand for imports will lead to a fall in the price and hence an improvement in the terms of trade. A decrease in the demand for exports will lead to a fall in the price and hence a deterioration in the terms of trade. An increase in the demand for imports will lead to a rise in the price and hence a deterioration in the terms of trade.

Supply factors
An increase in the supply of imports will lead to a fall in the price and hence an improvement in the terms of trade. A decrease in the supply of exports will lead to a rise in the price and hence an improvement in the terms of trade. A decrease in the supply of imports will lead to a rise in the price and hence a deterioration in the terms of trade. An increase in the supply of exports will lead to a fall in the price and hence a deterioration in the terms of trade.

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Exchange rate
A rise in the exchange rate of domestic currency will lead to a fall in the price of imports and hence an improvement in the terms of trade. A fall in the exchange rate of domestic currency will lead to a rise in the price of imports and hence a deterioration in the terms of trade.

4.2 Effect of a change in the terms of trade on the balance of trade

The effect of a change in the terms of trade on the balance of trade depends on the cause of the change.

4.2.1 Demand factors

When there is a change in the demand for exports, the price and the quantity will change in the same direction. For example, an increase in the demand for exports will lead to a rise in both the price and the quantity. When this happens, both the terms of trade and the balance of trade will improve.

When there is a change in the demand for imports, the price and the quantity will change in the same direction. For example, an increase in the demand for imports will lead to a rise in both the price and the quantity. When this happens, both the terms of trade and the balance of trade will deteriorate.

Therefore, when there is a change in the demand for exports or imports, the terms of trade and the balance of trade will change in the same direction.

Note: Although the terms of trade are affected by the change in the price, the balance of trade is affected by the change in the price and the change in the quantity.

4.2.2 Supply factors

When that is a change in the supply of exports, the price and the quantity will change in opposite directions. For example, an increase in the supply of exports will lead to a fall in the price and a rise in the quantity. When this happens, although the terms of trade will deteriorate, the balance of trade will depend on the price elasticity of demand for exports. If the demand for exports is price inelastic, which means that the decrease in the price will lead to a smaller proportionate increase in the quantity demanded, the balance of trade will also deteriorate. However, if the demand for exports is price elastic, which means that the decrease in the price will lead to a larger proportionate increase in the quantity demanded, the balance of trade will improve.

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When there is a change in the supply of imports, the price and the quantity will change in opposite directions. For example, an increase in the supply of imports will lead to a fall in the price and a rise in the quantity. When this happens, although the terms of trade will improve, the balance of trade will depend on the price elasticity of demand for imports. If the demand for imports is price inelastic, which means that the decrease in the price will lead to a smaller proportionate increase in the quantity demanded, the balance of trade will also improve. However, if the demand for imports is price elastic, which means that the decrease in the price will lead to a larger proportionate increase in the quantity demanded, the balance of trade will deteriorate.

Therefore, when there is a change in the supply of exports (imports), the terms of trade and the balance of trade will change in the same direction if the demand for exports (imports) is price inelastic. However, if the demand for exports (imports) is price elastic, the terms of trade and the balance of trade will change in opposite directions.

4.2.3 Exchange rate

When there is a rise in the exchange rate of domestic currency, which will lead to an improvement in the terms of trade, whether the balance of trade will improve or deteriorate will depend on the sum of the price elasticity of demand for exports and the price elasticity of demand for imports. If the sum is less than one, which means that the Marshall-Lerner condition does not hold, the balance of trade will improve. However, if the sum is greater than one, which means that the Marshall-Lerner condition holds, the balance of trade will deteriorate.

When there is a fall in the exchange rate of domestic currency, which will lead to a deterioration in the terms of trade, whether the balance of trade will improve or deteriorate will depend on the sum of the price elasticity of demand for exports and the price elasticity of demand for imports. If the sum is less than one, which means that the Marshall-Lerner condition does not hold, the balance of trade will deteriorate. However, if the sum is greater than one, which means that the Marshall-Lerner condition holds, the balance of trade will improve.

Therefore, when there is a change in the exchange rate of domestic currency, the terms of trade and the balance of trade will change in the same direction if the Marshall-Lerner condition does not hold. However, if the Marshall-Lerner condition holds, the terms of trade and the balance of trade will change in opposite directions.

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4.3 Marshall-Lerner condition

A fall in the exchange rate of domestic currency will lead to a rise in the price of imports. When the price of imports rises, the quantity demanded will fall. If the demand for imports is price elastic, which means that the increase in the price will lead to a larger proportionate decrease in the quantity demanded, import expenditure will decrease which will lead to an improvement in the balance of trade. If the demand for imports is price inelastic, import expenditure will rise. However, the balance of trade may not deteriorate. A fall in the exchange rate of domestic currency will reduce the price of exports which will lead to an increase in the quantity demanded. Since the price of exports in domestic currency will not be affected by a fall in the exchange rate of domestic currency, an increase in the quantity demanded will lead to an increase in export revenue. Therefore, if the sum of the price elasticity of demand for exports and the price elasticity of demand for imports is greater than one, which means that the Marshall-Lerner condition holds, the increase in export revenue will be greater than the increase in import expenditure which will lead to an improvement in the balance of trade, assuming export revenue is equal to import expenditure initially. However, if the Marshall-Lerner condition does not hold, the increase in export revenue will be less than the increase in import expenditure which will lead to a deterioration in the balance of trade, assuming export revenue is equal to import expenditure initially.

Let
P_X be the price of exports in foreign currency,
P_M be the price of imports in domestic currency,
P_X^{DC} be the price of exports in domestic currency,
$|PED_M|$ be the price elasticity of demand for imports,
$|PED_X|$ be the price elasticity of demand for exports,
BOT be the balance of trade and
E be the exchange rate of domestic currency.

Assume that
$|PED_M| = 0.6$,
$BOT = P_X^{DC}Q_X - P_MQ_M = 0$ (i.e. $P_X^{DC}Q_X = P_MQ_M$) and
$E\downarrow = 10\%$.

$E\downarrow(10\%) \Rightarrow P_M\uparrow(10\%) \Rightarrow Q_M\downarrow(6\%) \Rightarrow P_MQ_M\uparrow(4\%)$

Since P_MQ_M increases by 4%, the BOT will improve if $P_X^{DC}Q_X$ increases by more than 4%.

$E\downarrow(10\%) \Rightarrow P_X\downarrow(10\%) \Rightarrow Q_X\uparrow$

Since a fall in E will not affect P_X^{DC}, $P_X^{DC}Q_X$ will increase by more than 4% if Q_X increases by more than 4%, which means that $|PED_X| > 0.4$. Therefore, a fall in E will improve the BOT if $|PEDX + PEDM| > 1$ (i.e. the Marshall-Lerner condition holds), assuming export revenue is equal to import expenditure initially.

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5 FREE TRADE AGRREMENT: SINGAPORE PERSPECTIVE

A free trade agreement (FTA) is an agreement between two or more economies to remove or reduce barriers to trade, such as tariffs, with the objective of increasing the cross-border movement of goods and services between the economies.

If Singapore signs more FTAs, the balance of payments may improve. The balance of payments is a record of all the transactions between the residents of the economy and the rest of the world over a period of time and is made up of the current account and the capital and financial account. Signing FTAs will make Singapore's goods relatively cheaper than foreign goods in the FTA member countries due to the removal or reduction in tariffs on Singapore's goods. Therefore, if Singapore signs more FTAs, the exports to the FTA member countries will increase which will lead to an improvement in the current account and hence the balance of payments. Further, firms from non-member countries that export goods to the FTA member countries will increase investments in Singapore to circumvent the tariffs imposed on their goods in the FTA member countries. In doing so, their goods produced in Singapore and exported to the FTA member countries will become relatively cheaper than foreign goods which will lead to an increase in their sales and hence profits. The resultant increase in foreign direct investments in Singapore will lead to an improvement in the capital and financial account and hence the balance of payments.

Signing more FTAs may lead to an increase in aggregate demand and hence national income in Singapore. Aggregate demand is the total demand for the goods and services produced in the economy over a period of time and is comprised of consumption expenditure, investment expenditure, government expenditure on goods and services and net exports. The increase in exports and investment expenditure in Singapore will lead to an increase in the aggregate demand and hence the national income.

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In the above diagram, an increase in aggregate demand (AD) from AD_0 to AD_1 leads to an increase in national income (Y) from Y_0 to Y_1. When aggregate demand rises, firms will employ more factor inputs to produce more output and hence pay more factor income to households. Household income and hence consumption expenditure will rise. Due to the increase in consumption expenditure, firms will employ even more factor inputs to produce even more output and hence pay even more factor income to households. Household income and hence consumption expenditure will rise further. Therefore, an increase in aggregate demand will lead to a larger increase in national income and this is commonly known as the multiplier effect.

Since national income is equal to national output, the increase in national income in Singapore will lead to a rise in the demand for labour resulting in a fall in the unemployment, assuming the size of the labour force remains the same.

If Singapore signs more FTAs, the aggregate supply will rise. Aggregate supply is the total supply of goods and services in the economy over a period of time. Signing more FTAs will lead to a fall in the prices of imported intermediate goods in Singapore due to the removal or reduction in tariffs on the goods. Therefore, the cost of production in Singapore will fall which will lead to an increase in the aggregate supply. An increase in aggregate supply in Singapore will lead to an increase in the national income and hence a fall in the unemployment. Assuming aggregate demand in Singapore is rising, which is the normal state of the economy, an increase in the aggregate supply will also lead to a smaller rise in the general price level, and if this makes Singapore's goods and services relatively cheaper than foreign goods and services, net exports in Singapore will rise which will lead to an improvement in the current account and hence the balance of payments.

Signing more FTAs will lead to a more rapid increase in aggregate supply in Singapore in the long run. The increase in investment expenditure in Singapore will lead to a more rapid increase in the production capacity in the long run, assuming the net investment is initially positive. Therefore, aggregate supply in Singapore will rise more rapidly in the long run. When this happens, assuming aggregate demand in Singapore is rising, which is the normal state of the economy, the national income will rise more rapidly which will reduce the unemployment, and the general price level will rise less rapidly which may improve the balance of payments.

Although signing more FTAs will bring about beneficial effects to the Singapore economy, it will also bring about detrimental effects. Signing FTAs will make the FTA member countries' goods relatively cheaper than Singapore's goods in Singapore. Therefore, if Singapore signs more FTAs, the imports from the FTA member countries will increase. When national income in Singapore rises, the imports will rise. Further, the rise in the general price level in Singapore will make Singapore's goods and services relatively more expensive than foreign goods and services resulting in a decrease in the net exports. The increase in foreign direct investments in Singapore will also lead to an increase in the outward income remittances. When these happen, the current account and hence the balance of payments of Singapore will deteriorate. Cheaper imports from the FTA member countries in Singapore due to the removal or reduction in tariffs on the goods may induce

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households in Singapore to reduce consumption of domestic goods. If this happens, aggregate demand and hence national income in Singapore will fall which will lead to a rise in the unemployment. Further, foreign firms are footloose and hence if market conditions in other economies become more favourable in the future, they may pull their operations out of Singapore. If this happens, unemployment in Singapore may rise sharply. Signing more FTAs may also lead to a more rapid decline in the low value-added industries in Singapore, especially if the FTA partners are developing economies where Singapore imports most of its low value-added goods from. If this happens, structural unemployment in Singapore will rise. Since signing more FTAs will increase exports and imports in Singapore, the Singapore economy will become more susceptible to adverse economic conditions in other economies, such as recession and high inflation. As the exports and imports of Singapore are already very high, this may cause the economy to become rather unstable. Further, if Singapore signs more FTAs, some of the infant industries may not survive, especially if the FTA partners are developed economies where many of the high value-added goods are produced in mature industries. If this happens, the number of goods of comparative advantage and hence the range of goods produced in Singapore will decrease in the long run which may also cause the economy to become rather unstable. If Singapore signs more FTAs, the more rapid expansion of the export industries that produce high value-added goods and the more rapid decline in the low value-added industries will cause the income inequity to worsen.

6 THE GROWTH OF CHINA: SINGAPORE PERSPECTIVE

The growth of the Chinese economy may lead to a deterioration in the balance of payments of Singapore. The balance of payments is a record of all the transactions between the residents of the economy and the rest of the world over a period of time and is made up of the current account and the capital and financial account. Due to its larger pool of low-skilled labour, China has a comparative advantage over Singapore in producing low value-added goods, which include consumer, capital and intermediate goods. Therefore, the growth of the Chinese economy will lead to a fall in Singapore's exports of low value-added goods. When this happens, the current account and hence the balance of payments of Singapore will deteriorate. When the Chinese economy grows, China will attract some foreign direct investments from Singapore due to its lower labour cost and larger consumer market. Further, due to the same reasons, firms in Singapore will increase investments in China. When this happens, the capital and financial account and hence the balance of payments of Singapore will deteriorate.

When the Chinese economy grows, aggregate demand and hence national income in Singapore may fall. Aggregate demand is the total demand for the goods and services produced in the economy over a period of time and is comprised of consumption expenditure, investment expenditure, government expenditure on goods and services and net exports. Due to the decrease in exports and investment expenditure in Singapore, the aggregate demand and hence the national income will fall.

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In the above diagram, a decrease in aggregate demand (AD) from AD_0 to AD_1 leads to a decrease in national income (Y) from Y_0 to Y_1. When aggregate demand falls, firms will employ less factor inputs to produce less output and hence pay less factor income to households. Household income and hence consumption expenditure will decrease. Due to the decrease in consumption expenditure, firms will employ even less factor inputs to produce even less output and hence pay even less factor income to households. Household income and hence consumption expenditure will decrease further. Therefore, a decrease in aggregate demand will lead to a larger decrease in national income and this is commonly known as the reverse multiplier effect.

Since national income is equal to national output, the decrease in national income in Singapore will lead to a fall in the demand for labour resulting in a rise in the unemployment, assuming the size of the labour force remains the same.

The decrease in aggregate demand in Singapore will lead to a decrease in the demand and hence the prices of factor inputs resulting in a fall in the cost of production and hence the general price level. In the above diagram, a decrease in aggregate demand (AD) from AD_0 to AD_1 leads to a fall in the general price level (P) from P_0 to P_1. When the general price level in Singapore falls, people may expect it to fall further. If this happens, consumption expenditure in Singapore will fall which will lead to a further decrease in the aggregate demand.

The growth of the Chinese economy may lead to a decrease in aggregate supply in Singapore. Aggregate supply is the total supply of goods and services in the economy over a period of time. When the Chinese economy grows, the world demand for energy and hence energy prices will rise. Further, China will experience inflation and hence the prices of imported intermediate goods from China will also rise. Therefore, the cost of production in Singapore will rise which will lead to a decrease in the aggregate supply. A decrease in aggregate supply in Singapore will lead to a decrease in the national income and hence a

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rise in the unemployment. A decrease in aggregate supply in Singapore will also lead to a shortage of goods and services resulting in a rise in the general price level, and if this makes Singapore's goods and services relatively more expensive than foreign goods and services, net exports in Singapore will fall which will lead to a deterioration in the current account and hence the balance of payments.

Due to China's comparative advantage over Singapore in producing low value-added goods, the growth of the Chinese economy will cause the low value-added industries in Singapore to decline more rapidly which will lead to a rise in the structural unemployment.

The more rapid decline in the low value-added industries in Singapore will depress the demand for low-skilled labour and hence the wages which will cause the income inequity to worsen.

When the Chinese economy grows, aggregate supply in Singapore may rise less rapidly in the long run. The decrease in investment expenditure in Singapore will lead to a less rapid increase in the production capacity in the long run, assuming the net investment remains positive. Therefore, aggregate supply in Singapore will rise less rapidly in the long run. When this happens, assuming aggregate demand in Singapore is rising, which is the normal state of the economy, the national income will rise less rapidly which may increase the unemployment, and the general price level will rise more rapidly which may worsen the balance of payments.

Although the growth of the Chinese economy will bring about detrimental effects to the Singapore economy, it will also bring about beneficial effects. When the Chinese economy grows, the people will become more affluent and hence their demand for high value-added consumer goods will rise. Further, to feed its economic growth, China will need more high-value-added capital and intermediate goods. Due to its larger pool of high-skilled labour, Singapore has a comparative advantage over China in producing high value-added goods, which include consumer, capital and intermediate goods. Therefore, the growth of the Chinese economy will lead to a rise in Singapore's exports of high value-added goods. The increase in outward foreign direct investments in Singapore will also lead to an increase in the inward income remittances. When these happen, the current account and hence the balance of payments of Singapore will improve. When the Chinese economy grows, some of the firms which will become larger will expand to other economies including Singapore. When this happens, the capital and financial account and hence the balance of payments of Singapore will improve. Due to the increase in exports and investment expenditure in Singapore, the aggregate demand will rise which will lead to an increase in the national income resulting in a fall in the unemployment. When the Chinese economy grows, it will produce more of the intermediate goods that Singapore needs. Therefore, Singapore's imports of intermediate goods from China will rise. Due to the lower prices, the cost of production in Singapore will fall which will lead to an increase in the aggregate supply resulting in an increase in the national income and hence a fall in the unemployment. The increase in investment expenditure in Singapore will lead to a more rapid increase in the production capacity and hence the aggregate supply in the long run, assuming the net investment is initially positive. When this happens, assuming aggregate

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demand in Singapore is rising, which is the normal state of the economy, the national income will rise more rapidly which may decrease the unemployment, and the general price level will rise less rapidly which may improve the balance of payments.

7 GLOBALISATION: DEVELOPING ECONOMIES VERSUS DEVELOPED ECONOMIES

Globalisation refers to the increase in flows of goods, services, investments and labour across international borders.

Globalisation may lead to an improvement in the balance of payments of developing economies. The balance of payments is a record of all the transactions between the residents of the economy and the rest of the world over a period of time and is made up of the current account and the capital and financial account. Due to their larger pool of low-skilled labour, developing economies have a comparative advantage over developed economies in producing low value-added goods, which include consumer, capital and intermediate goods. Therefore, globalisation will lead to an increase in exports of low value-added goods in developing economies. When this happens, their current account and hence balance of payments will improve. Further, globalisation will lead to a flow of foreign direct investments from developed economies to developing economies due to the lower labour cost in developing economies. When this happens, the capital and financial account of developing economies will improve.

Aggregate demand and hence national income in developing economies may rise due to globalisation. Aggregate demand is the total demand for the goods and services produced in the economy over a period of time and is comprised of consumption expenditure, investment expenditure, government expenditure on goods and services and net exports. Due to the increase in exports and investment expenditure in developing economies, the aggregate demand and hence the national income will rise.

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In the above diagram, an increase in aggregate demand (AD) from AD_0 to AD_1 leads to an increase in national income (Y) from Y_0 to Y_1. When aggregate demand rises, firms will employ more factor inputs to produce more output and hence pay more factor income to households. Household income and hence consumption expenditure will increase. Due to the increase in consumption expenditure, firms will employ even more factor inputs to produce even more output and hence pay even more factor income to households. Household income and hence consumption expenditure will increase further. Therefore, an increase in aggregate demand will lead to a larger increase in national income and this is commonly known as the multiplier effect.

Since national income is equal to national output, the increase in national income in developing economies will lead to a rise in the demand for labour resulting in a fall in the unemployment, assuming the size of the labour force remains the same.

Globalisation may lead to a more rapid increase in aggregate supply in developing economies in the long run. Aggregate supply is the total supply of goods and services in the economy over a period of time. The import of better production technologies from developed economies and the increase in investment expenditure in developing economies will lead to a more rapid increase in the production capacity in the long run, assuming the net investment is initially positive. Therefore, aggregate supply in developing economies will rise more rapidly in the long run. When this happens, assuming aggregate demand in developing economies is rising, the national income will rise more rapidly which may decrease the unemployment, and the general price level will rise less rapidly which may improve the balance of payments.

Globalisation may bring about detrimental effects to developed economies. Due to the same reasons that may lead to an improvement in the balance of payments of developing economies, an increase in the aggregate demand and a more rapid increase in the aggregate supply in the long run, the balance of payments of developed economies may worsen, the aggregate demand may fall and the aggregate supply may rise less rapidly in the long run. Due to developing economies' comparative advantage over developed economies in producing low value-added goods, globalisation will cause the low value-added industries in developed economies to decline more rapidly which will lead to a rise in the structural unemployment. The more rapid decline in the low value-added industries in developed economies will depress the demand for low-skilled labour and hence the wages which will cause the income inequity to worsen.

Although developing economies can benefit more from globalisation than developed economies, this is not necessarily true.

Globalisation may lead to an improvement in the balance of payments of developed economies. When developing economies grow, the people will become more affluent and hence their demand for high value-added consumer goods will rise. Further, to feed its economic growth, developing economies will need more high value-added capital and intermediate goods. Due to their larger pool of high-skilled labour, developed economies have a comparative advantage over developing economies in producing high value-added

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goods, which include consumer, capital and intermediate goods. Therefore, the growth of developing economies will lead to an increase in exports of high value-added goods in developed economies. The increase in outward foreign direct investments in developed economies will also lead to an increase in the inward income remittances. When these happen, the current account and hence the balance of payments of developed economies will improve. When developing economies grow, some of the firms which will become larger will expand to other economies including developed economies. When this happens, the capital and financial account and hence the balance of payments of developed economies will improve.

The increase in exports and investment expenditure in developed economies will lead to an increase in the aggregate demand and hence the national income resulting in a fall in the unemployment.

Aggregate supply in developed economies may rise due to globalisation. When developing economies grow, they will produce more of the intermediate goods that developed economies need. Therefore, developed economies' imports of intermediate goods from developing economies will rise. Due to the lower prices, the cost of production in developed economies will fall which will lead to an increase in the aggregate supply. The cost of production in developed economies will also fall due to an increase in inflow of cheaper labour from developing economies, both high-skilled and low-skilled.

Globalisation may lead to a more rapid increase in aggregate supply in developed economies in the long run. The inflow of foreign labour and the increase in investment expenditure in developed economies will lead to a more rapid increase in the production capacity in the long run, assuming the net investment is initially positive. Therefore, aggregate supply in developed economies will rise more rapidly in the long run.

Globalisation may bring about detrimental effects to developing economies. Due to the same reasons that may lead to an improvement in the balance of payments of developed economies, the balance of payments of developing economies may worsen. The outflow of labour in developing economies may lead to a decrease in the supply of labour resulting in a decrease in the aggregate supply. The problem of brain drain in developing economies may make it difficult for them to move up the value-added chain which may lead to lower economic growth in the long run. The entry of multinational corporations into developing economies may lead to closure of the small firms that reap less economies of scale, and if this causes the economies to become over-dependent on these corporations that are footloose, the unemployment may rise sharply if they pull their operations out of the economies in the future due to more favourable market conditions in other economies. Due to the lax labour law and low environmental standards in developing economies, the entry of multinational corporations may lead to labour exploitation and environmental degradation which may lower the standard of living.

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8 GLOBALISATION: SINGAPORE PERSPECTIVE

Globalisation has led to an improvement in the balance of payments of Singapore. The balance of payments is a record of all the transactions between the residents of the economy and the rest of the world over a period of time and is made up of the current account and the capital and financial account. Globalisation has increased the imports of Singapore substantially over the last few decades. Singapore's imports are now close to 200 per cent of its national income. However, globalisation has increased the exports of Singapore by a larger amount in the same period. Singapore's exports are now over 200 per cent of its national income. The larger increase in exports than imports in Singapore has led to an improvement in the current account and hence the balance of payments. Further, although globalisation has led to a substantial increase in outward foreign direct investments in Singapore, it has led to a larger increase in the inward foreign direct investments which has resulted in an improvement in the capital and financial account and hence the balance of payments.

Aggregate demand and hence national income in Singapore have risen due to globalisation. Aggregate demand is the total demand for the goods and services produced in the economy over a period of time and is comprised of consumption expenditure, investment expenditure, government expenditure on goods and services and net exports. Due to little overlap between the goods that Singapore produces and those that it imports, the increase in imports in Singapore has not led to a significant decrease in the aggregate demand. Therefore, the increase in exports and investment expenditure in Singapore has led to an increase in the aggregate demand and hence the national income.

In the above diagram, an increase in aggregate demand (AD) from AD_0 to AD_1 leads to an increase in national income (Y) from Y_0 to Y_1. When aggregate demand rises, firms will employ more factor inputs to produce more output and hence pay more factor income to households. Household income and hence consumption expenditure will increase. Due to the increase in consumption expenditure, firms will employ even more factor inputs to

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produce even more output and hence pay even more factor income to households. Household income and hence consumption expenditure will increase further. Therefore, an increase in aggregate demand will lead to a larger increase in national income and this is commonly known as the multiplier effect.

Since national income is equal to national output, the increase in national income in Singapore has generated many jobs for the expanding labour force which has led to lower unemployment in the economy.

The demand-side benefits of globalisation are greater in Singapore than in many other economies. Singapore has a small domestic sector and hence is rather export-dependent. Therefore, the increase in exports in Singapore due to globalisation has led to a substantial increase in the aggregate demand. Further, domestic firms in Singapore are small and hence do not have the financial resources to make large investments. Therefore, the increase in foreign direct investments made by multinational corporations in Singapore due to globalisation has also increased the aggregate demand substantially. Large economies like the United States, by contrast, depend more on the domestic sector and have large firms to make large investments and hence have benefited from the increase in exports and foreign direct investments due to globalisation to a smaller extent.

Globalisation has led to an increase in aggregate supply in Singapore. Aggregate supply is the total supply of goods and services in the economy over a period of time. Globalisation has led to an increase in the amount of imported intermediate goods and a rise in the number of immigrants and foreign workers in Singapore which has resulted in a more rapid increase in the production capacity and hence the aggregate supply. Further, the increase in foreign direct investments in Singapore has also led to a more rapid increase in the production capacity and hence the aggregate supply. Due to the increase in aggregate supply in Singapore, the national income has risen by a larger amount, the unemployment is lower and the general price level has risen by a smaller amount.

The supply-side benefits of globalisation are greater in Singapore than in many other economies. Singapore virtually does not have factor endowments. Therefore, the increase in the amount of imported intermediate goods in Singapore due to globalisation has increased the aggregate supply substantially. Further, Singapore has a small labour force due to the small population. Therefore, the rise in the number of foreign workers in Singapore due to globalisation has also increased the aggregate supply substantially. Large economies like Japan, by contrast, have more factor endowments and a larger labour force and hence have benefited from the increase in the amount of imported intermediate goods and the rise in the number of foreign workers due to globalisation to a smaller extent.

Although Singapore has benefited more from globalisation than many other economies, it has also suffered more from globalisation than many other economies which may have resulted in a lower net benefit.

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Globalisation has increased the susceptibility of the Singapore economy to a recession in other economies. Given that Singapore's exports are now over 200 per cent of its national income, a recession in other economies will lead to a large decrease in the external demand for Singapore's goods and services. When this happens, aggregate demand and hence national income in Singapore will fall substantially.

Globalisation has increased the susceptibility of the Singapore economy to high inflation in other economies. Given that Singapore's imports are now close to 200 per cent of its national income, high inflation in other economies will lead to a rapid rise in the prices of imported goods and services in Singapore, which include both consumer and intermediate goods, resulting in high imported inflation in the economy. The rapid rise in the prices of imported intermediate goods in Singapore will also lead to high cost-push inflation in the economy.

Globalisation has led to a more rapid change in the structure of the Singapore economy and hence a more rapid decline in the low value-added industries in Singapore resulting in a rise in the structural unemployment.

Globalisation has worsened income inequity in Singapore. Globalisation has led to a more rapid increase in the external demand of the Singapore economy than the domestic demand. As a result, the profits of firms in Singapore that produce goods for export have been rising more rapidly than the profits of the firms that produce goods for the domestic market causing the income inequity to worsen. Further, the wages of high-skilled workers in Singapore have been rising more rapidly than the wages of the low-skilled workers due to the more rapid expansion of the export industries that produce high value-added goods and the more rapid decline in the low value-added industries and this has also caused the income inequity to worsen.

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